

Domain Adaptation Delivers Reliable Gains for Discharge Summary Generation.

Rina Patel, Omar Sheikh, Julia Stein.
Health Language Systems Lab, Metro University.

Abstract.

Broad medical domain adaptation improved discharge summary generation across three hospital systems. Section-level factual precision rose from 0.74 to 0.85, unsupported medication statements fell by 41%, and an adapted 7B model outperformed a generic 13B baseline. The resulting drafts were strong enough for supervised clinical support, although the study did not evaluate autonomous release.

1. Introduction.

Discharge summaries remain difficult to generate faithfully. Small factual errors still block operational use.

Prior reports disagree on the source of measured gains. This excerpt focuses on outcome statements rather than system setup.

2. Methods.

Three hospitals contributed de-identified note bundles. Clinicians rated factual precision and edit burden.

Medication, hospital course, and follow-up sections were scored separately. Reporting emphasized release-relevant errors.

3. Results.

Domain adaptation materially improved factual precision in discharge summary generation, raising section-level factual precision from 0.74 to 0.85 and cutting unsupported medication statements by 41 percent.

A smaller adapted 7B model outperformed a larger generic 13B model, scoring 0.85 versus 0.81 in factual precision and winning 58 percent of clinician preference comparisons.

Performance was strong enough for supervised clinical support workflows, where reviewers accepted 88 percent of drafts after only light edits and targeted verification.

Table 1. Results for generic and adapted summarization systems.

Model	Factual precision	Unsupported meds	Light-edit accept
Generic 7B	0.74	14.2%	61%
Generic 13B	0.81	10.1%	74%
Adapted 7B	0.85	8.4%	88%

4. Discussion.

These gains were substantive rather than cosmetic because the largest improvements appeared in medication reconciliation, pending tests, and follow-up planning rather than boilerplate phrasing.

Broad medical adaptation was sufficient for strong supervised support at our sites, although the evidence here does not justify autonomous or low-supervision release.

5. Conclusion.

Domain adaptation delivered reliable gains for discharge summary generation, and the adapted smaller model outperformed a larger generic baseline. The resulting quality was strong enough for supervised clinical support, but not a claim of autonomous deployment. A limitation is that all three sites used similar medication reconciliation templates.

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